Ali Khoramfar

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Research Interests

- Natural Language Processing • Vision-Language Models • Multimodal Learning

Education

University of Tehran *2023 – Present*

Master of Science in Computer Engineering

- **GPA: 19.33/20** (4/4)
- **Relevant Courses:** Natural Language Processing, Deep Learning, Machine Learning, Social Networks, Statistical Inference, Distributed Machine Learning
- **Master's Thesis** (Supervisors: Prof. Heshaam Faili - Prof. Mohammad Javad Dousti):
  - **Title:** Investigating the Performance of Multimodal Large Language Models for Medical Image Analysis
  - **Summary:** Analyzing the performance of multimodal large language models on complex reasoning tasks and multilingual settings, with medical image analysis employed as a representative application.

JundiShapur University of Technology *2017 – 2022*

Bachelor of Science in Computer Engineering

- **GPA: 19.11/20** (4/4)
- **Ranked 1st** among all Computer Engineering students
- **Relevant Courses:** Artificial Intelligence and Expert Systems, Image Processing, Internet Engineering, Microprocessors, Principles of Database Design
- **Bachelor's Project** (Supervisor: Dr. Maryam Chinipardaz):
  - **Title:** Green Internet of Things and Solar Energy
  - **Summary:** Designed a five-layer architecture to optimize IoT energy efficiency and investigated solar-powered solutions for sustainable deployment.

Publications

Conference Papers

- *PerMed-MM: A Multimodal, Multi-Specialty Persian Medical Benchmark for Evaluating Vision Language Models*
  Khoramfar A., Dousti M. J., & Faili H. **IJCNLP-AACL 2025 (Main Conference, Poster Presentation)**
- *Stable Evidence, Unstable Decisions: An Empirical Analysis of Model Decision Stability in VLMs*
  Khoramfar A., Dousti M. J., Mohamadian A., & Faili H. **(Under Review at ACL 2026)**
- *DeepQuestion: Systematic Generation of Real-World Challenges for Evaluating LLM Performance*
  Khoramfar A., Ramezani A., Mohajeri M. M., Dousti M. J., Nili Ahmadabadi M., & Faili H. **(Under Review at LREC 2026)**
- *PlantSee: Cross-Lingual Adaptation of Vision–Language Models for Plant Disease Diagnosis*
  Ashouri Sefat S., Khoramfar A., & Amini S. **(Under Review at LREC 2026)**

Journal Papers

- *Green Internet of Things and Solar Energy*
  Chinipardaz, M., Khoramfar, A., & Amraee, S. (2024). Environmental Science and Pollution Research, 31(12), 18296–18312. [DOI]

Other Publications

- *Network Level Energy Solutions for Green Internet of Things*
  Khoramfar A., & Chinipardaz M. (2023). First National Conference on Internet of Things
- *Green Computing Solutions in Environmental Protection and Use of Renewable Energy*
  Amraei S., & Khoramfar A. (2021). National Conference on Chemical Engineering & Nanotechnology (in Persian)
- *The Role of IoT in Innovation and Business Development*
  Khoramfar A., & Chinipardaz M. (2021). First National Conference on Entrepreneurial Universities (in Persian)

Research Experiences

Graduate Student Researcher, University of Tehran *2023 – Present*

Natural Language Processing Laboratory

- Conducted research on multimodal and multilingual evaluation of LLMs and VLMs, focusing on reasoning, benchmark construction, and low-resource languages.
- Led dataset curation and large-scale experimental evaluations, contributing to multiple research papers.
- Reviewed submissions for ACL Rolling Review, providing detailed technical and methodological feedback.

Undergraduate Research Assistant, JundiShapur University *2020 – 2022*

Computer Networks Laboratory

- Engaged in research on energy-efficient computing.
- Conducted literature reviews and collaborated on writing and revising research manuscripts.

Teaching Experiences

Distributed Machine Learning – Head TA *Sep 2025 – Present*

Supervisor: Prof. Mohammad Javad Dousti

- Designed assignments focusing on scalability, load balancing, and communication–computation trade-offs in distributed machine learning systems.

Essential Computing Skills – Head TA *Jan 2025 – Sep 2025*

Supervisor: Prof. Mohammad Javad Dousti

- Coordinated course logistics, including schedule management and preparation of course materials.
- Assisted with grading and facilitated technical workshops on Vim, Git, Docker and Kubernetes.

Natural Language Processing – TA *Aug 2024 – Feb 2025*

Supervisor: Prof. Heshaam Faili

- Assisted in assessment design and grading and developed a project on Vision-Language Models (CLIP), introducing students to multimodal embeddings, similarity analysis and bias evaluation in pre-trained models.

Deep Learning – TA *Aug 2024 – Feb 2025*

Supervisor: Dr. Ahmad Kalhor

- Designed and supervised a course project introducing students to Vision Transformers, including data preprocessing, transfer learning, fine-tuning strategies and comparative evaluation with CNN architectures.

Internet Engineering – TA *Sep 2021 – Jan 2022*

Supervisor: Dr. Maryam Chinipardaz

- Delivered technical workshops on web hosting and search engines, covering hosting configuration, domain management, deployment, indexing, ranking algorithms and SEO optimization practices.

Academic Projects

Distributed Machine Learning [GitHub] *Sep 2024 – Jan 2025*

Supervisor: Prof. Mohammad Javad Dousti

Natural Language Processing [GitHub] *Jan 2024 – Jul 2024*

Supervisor: Prof. Heshaam Faili

Deep Learning [GitHub] *Jan 2024 – Jul 2024*

Supervisor: Dr. Ahmad Kalhor

Machine Learning [GitHub] *Jan 2024 – Jul 2024*

Supervisor: Dr. Tavassolipour

Statistical Inference [GitHub] *Sep 2023 – Jan 2024*

Supervisor: Dr. A. Dehaqani

Technical Skills

Programming: Python (NumPy, Pandas, Matplotlib, scikit-learn, PySpark), C++, C#, SQL
Frameworks: PyTorch Distributed, Hugging Face (Transformers, Datasets, PEFT), LangGraph, Unsloth
Tools: Git, MPI, Slurm

Languages

English: Professional Working Proficiency
Persian: Native